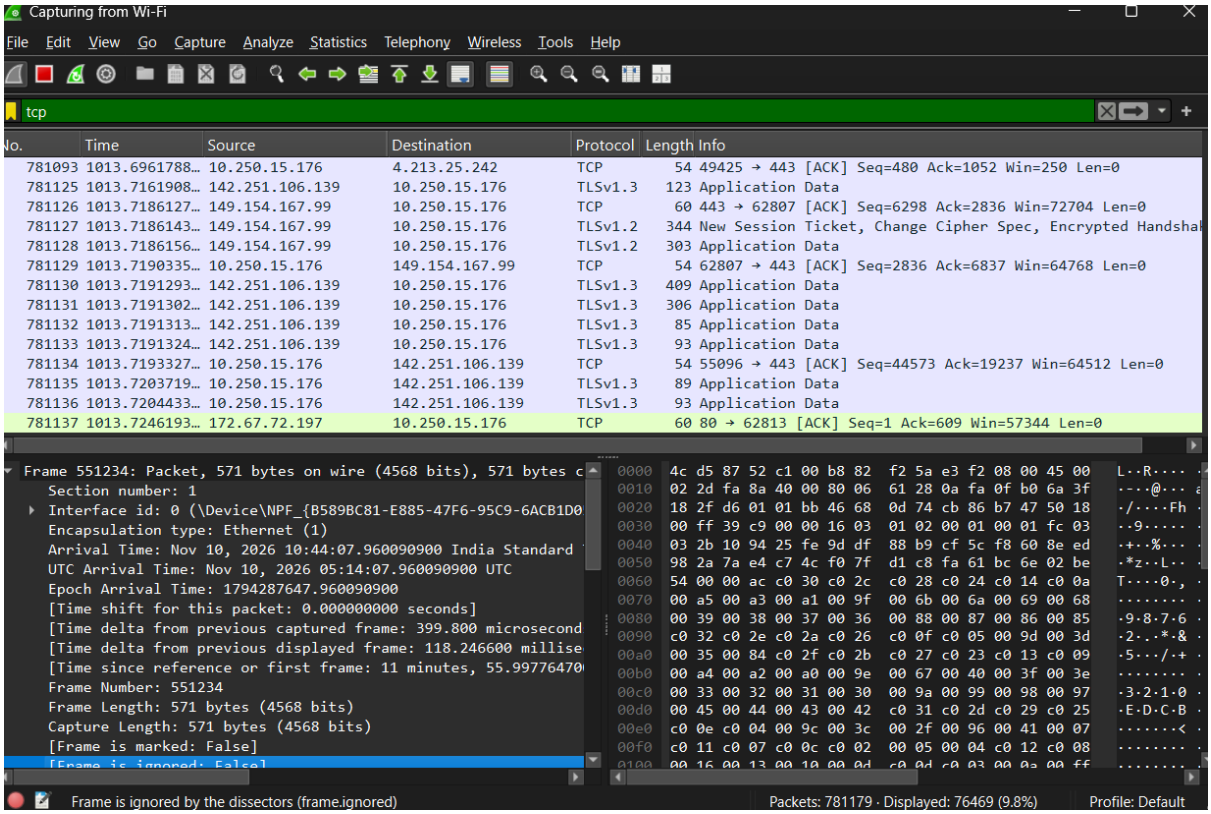
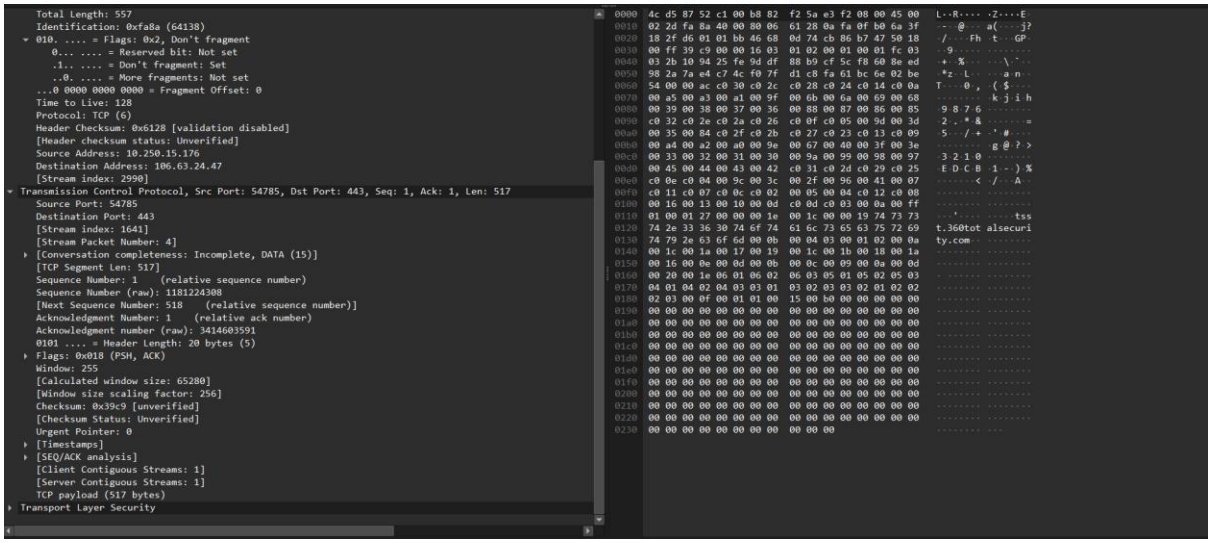
Skill - 1

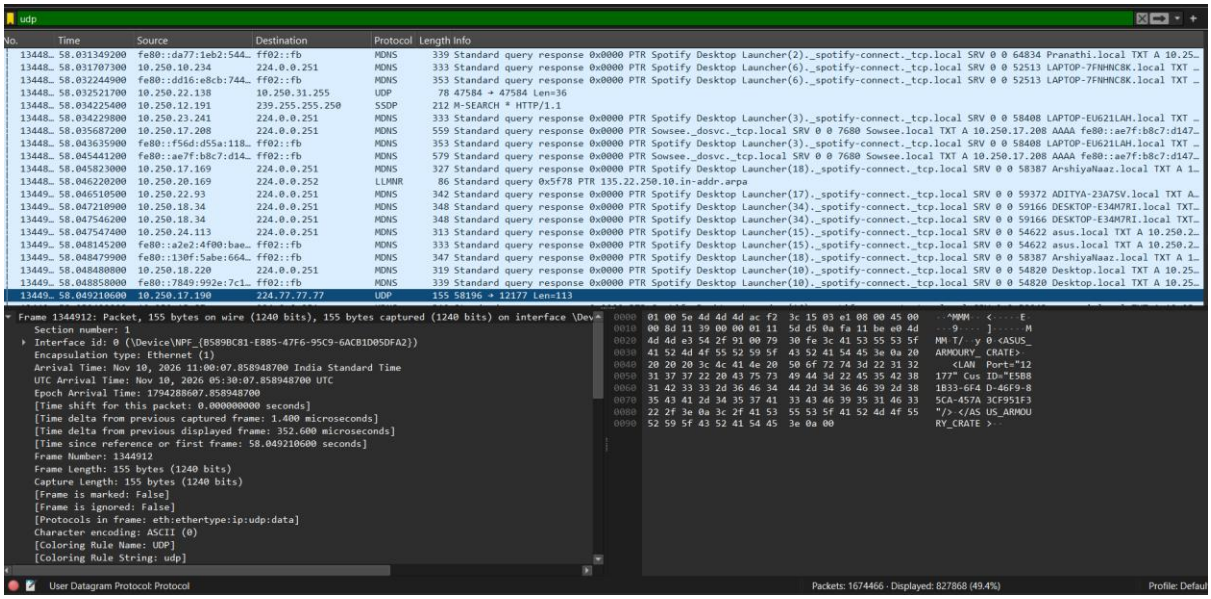
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

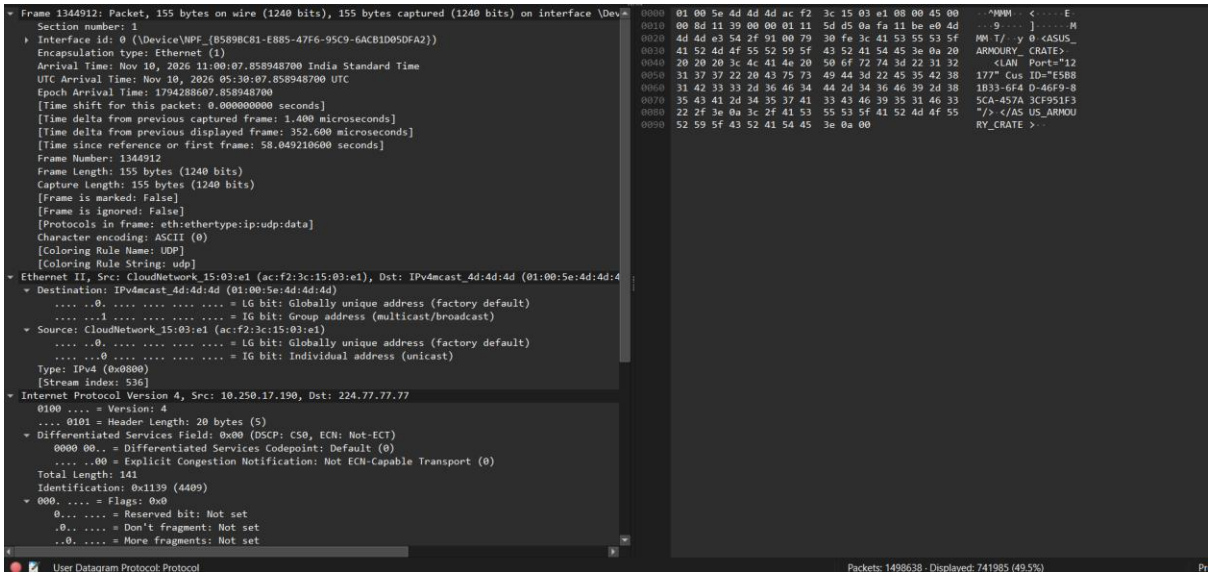


```
Frame 551234: Packet, 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface \D
Ethernet II, Src: WNC_5a:e3:f2 (b8:82:f2:5a:e3:f2), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
  Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: WNC_5a:e3:f2 (b8:82:f2:5a:e3:f2)
  Type: IPv4 (0x0800)
  [Stream index: 116]
Internet Protocol Version 4, Src: 10.250.15.176, Dst: 106.63.24.47
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
     0000 00.. = Differentiated Services Codepoint: Default (0)
     .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 557
  Identification: 0xfa8a (64138)
  010. .... = Flags: 0x2, Don't fragment
     0... .... = Reserved bit: Not set
     .1.. .... = Don't fragment: Set
     ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x6128 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.250.15.176
  Destination Address: 106.63.24.47
  [Stream index: 2990]
Transmission Control Protocol, Src Port: 54785, Dst Port: 443, Seq: 1, Ack: 1, Len: 517
  Source Port: 54785
  Destination Port: 443
  [Stream index: 1641]
  [Stream Packet Number: 4]
  [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 517]
  Sequence Number: 1    (relative sequence number)
  Sequence Number (raw): 1181224308
  [Next Sequence Number: 518    (relative sequence number)]
  Acknowledgment Number: 1    (relative ack number)
  Acknowledgment number (raw): 3414603591
  0101 .... = Header Length: 20 bytes (5)
```





Frame 1344912: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Dev
Section number: 1
Interface id: 0 (\Device\NPF_{B589BC81-E885-47F6-95C9-6ACB1D05DFA2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 10, 2026 11:00:07.858948700 India Standard Time
UTC Arrival Time: Nov 10, 2026 05:30:07.858948700 UTC
Epoch Arrival Time: 1794288607.858948700
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.400 microseconds]
[Time delta from previous displayed frame: 352.600 microseconds]
[Time since reference or first frame: 58.049210600 seconds]
Frame Number: 1344912
Frame Length: 155 bytes (1240 bits)
Capture Length: 155 bytes (1240 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: CloudNetwork_15:03:e1 (ac:f2:3c:15:03:e1), Dst: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4
Destination: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4d)
.... ..0. .... = LG bit: Globally unique address (factory default)
....1 .... = IG bit: Group address (multicast/broadcast)
Source: CloudNetwork_15:03:e1 (ac:f2:3c:15:03:e1)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 536]
Internet Protocol Version 4, Src: 10.250.17.190, Dst: 224.77.77.77
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 141
Identification: 0x1139 (4409)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set



Parameter	TCP Packet	UDP Packet
Frame Length	571 bytes	155
Ethernet Header Length	14	14
IP Header Length	20	20
Transport Header Length	20	8
IP Total Length	557	141
Transport Header Length	20	8
Payload Length	517	113

Answer these questions

1. What is encapsulation?
Encapsulation is the process of combining data and methods into a single unit and restricting direct access to the data.

2. What is the purpose of the Ethernet header?
It contains the source and destination MAC addresses and identifies the type of data being transmitted.

3. What information is stored in the IP header?
It stores source and destination IP addresses, packet length, TTL, and protocol information.

4. What is the difference between TCP and UDP headers?
TCP headers contain information for reliable data delivery, sequencing, and error control, whereas UDP headers are simpler and contain port numbers, length, and checksum.

5. Why does each protocol add a header?
Each protocol adds a header to provide necessary information for identifying, routing, and delivering data correctly.

6. What is packet overhead?
Packet overhead is the extra data added to a packet in the form of headers and trailers.

7. How does header overhead affect network efficiency?
More header overhead means more data is transmitted for control purposes, leaving less space for actual data and reducing network efficiency.